

Multidimensional Parameter Decomposition for Mechanistic Interpretability

*Multidimensional Parameter Decomposition
for Mechanistic Interpretability*

Master's thesis in Data Science

Course: SPDS800

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ABSTRACT

Some super cool abstract of all my findings ¹ Add your github and folder for wandb.

1 Introduction

Mechanistic interpretability seeks to reverse-engineer neural networks into a set of human-interpretable mechanisms. Different branches of the field pursue this goal with different focal points: Either through directions in activation space, computational subgraphs, or components of the network’s parameters.

However, both the atomic structure for decomposition and the decomposition itself remain open problems. [Sha+25] Whether a decomposition recovers the computational primitives of a neural network depends on what the ground-truth decomposition is taken to be. With models consisting of independent features, the decomposition is unambiguous, with one atom per feature. If the features are dependent or co-occur, the decomposition becomes underdetermined. Decompositions that split features across multiple atoms, risk losing the structural information, that gives each feature its meaning, while decompositions that group multiple features into a single atom, risk losing granularity by collapsing distinct features into a single unit.

For example, a network computing the volume of a cube can be decomposed into components corresponding to individual side lengths, or into a single component computing the volume directly. The former corresponds to a feature-level decomposition, with one atom per input direction, exposing internal structure but leaving grouping implicit, only recoverable through co-activation patterns. The latter corresponds to a structural-level decomposition, with one atom per computation, exposing the grouping directly but collapsing per side-length information.

Recently, a formal definition of irreducibility was introduced to disambiguate cases such as the cube example. ² [Eng+25] Features are irreducible, and thus constitute units of decomposition, if their joint distribution cannot be decomposed into independent factors or into a mixture containing a lower-dimensional component.

In terms of the decomposition itself, recent work focuses on decomposing parameters, as opposed the activations. These methods aim to recover components that are faithful to the original weights, sparse per input, and individually simple. One such approach, Stochastic Parameter Decomposition (SPD) [BBS25], factorizes networks into rank-one components with a learned per component importance function.

This work extends SPD by examining its behaviour on dependent features, in settings where the multi-dimensional feature hypothesis posits a feature-level or a structural-level unit ³ [Eng+25]. Additionally a reformulation of the importance function is proposed, to encourage that features that are manipulated jointly are grouped together, by computing importance relationally across components.

My findings

2 Related work

2.1 The linear representation hypothesis

A widespread view in mechanistic interpretability is the linear representation hypothesis: high-level concepts are encoded as directions in a network’s representation space [PCV24]. Together with the superposition hypothesis [Elh+22], it forms a central tenet of interpretability: each feature corresponds to a single direction, and a hidden state is a sparse linear combination of feature directions.

Recent work questions how strictly this view should be taken, distinguishing between a weak form of the hypothesis, that some features are linear, from a stronger form that all features are [Men24; Smi24]. Empirically multiple counterexamples to the stronger form have been proposed, include recurrent networks encoding token position by magnitude rather than direction [Cso+24], transformers representing cyclical concepts such as days of the week [Eng+25] or modular arithmetic [Nan+23] in irreducible low-dimensional subspaces.

¹Very important to add quantization somewhere. <https://arxiv.org/pdf/2303.13506>, new SPD paper in related work and discussion - particularly they also say something about their minimality loss, and lastly extend the transformer thing in the paper to be more clear of the architecture. Particularly no positional encoding?

²I got this definition wrong, we decompose f not the V ’s, we thus mustn’t show delta orthogonality, and the argument with that the V ’s can be linear combinations isn’t as important anymore. However we still have the same argument, that they can be low dimensional instead of rank 1. Secondly we need to state clearly that f is a function $f : d_{input} \rightarrow \mathbb{R}^{d_{fi}}$

³I mean feature-level, structure level is wrong under this view but it sounds good

2.2 Activation-space decomposition

Decomposing a network’s activations predominantly focuses on disentangling layer activations into interpretable features. Here, sparse autoencoders (SAE) are one of the most widely researched methods [Cun+23b; Bri+23], that learn an overcomplete dictionary that expresses each hidden state as a sparse combination of one-dimensional features. However, these features are not necessarily aligned with the network’s computation, due to feature absorption, inconsistencies across dictionary sizes, and a failure to capture feature geometry. [Cha+25; Lea+25; Men24].

Circuit discovery extends feature analysis to identify how a computation is processed by the network’s structure, by identifying minimal subgraphs required to perform a given task, without loss of performance. [Ola+20; Mar+25]. However, these methods generally do not account for the polysemantic and densely connected structure of neural networks, where the functional behaviour can be attributed to multiple distinct but functionally equivalent subgraphs [Mél+25].

2.3 Parameter-space decomposition

Recently a newer line of work aims to decompose the network’s parameters directly, instead of the activations. Transcoders [DCN24] represent an intermediate step that learns a sparse approximation of an MLP layer. Parameter decomposition methods go further, factorising the network’s weights directly, with the goal of recovering the network’s computational primitives.

Most directly relevant to this work is the description-length-driven line of Attribution-based Parameter Decomposition [Bra+25] and Stochastic Parameter Decomposition (SPD) [BBS25], which has recently been explored in small transformer models [CR25] and cross-layer clustering [Ses+26]. While APD decomposes a network into multidimensional components, the method is sensitive to hyperparameters, and does not scale beyond toy settings due to computational complexity. SPD addresses this by decomposing each weight matrix into rank-one subcomponents and learning a per-input importance function over them, at the cost of requiring post-hoc clustering to recover higher-rank mechanisms; both methods are detailed in Section 3.1. Other parameter-space approaches include local loss landscape decomposition [CBS25].

3 Method

3.1 Background on SPD

Attribution-based Parameter Decomposition (APD) [Bra+25] and Stochastic Parameter Decomposition (SPD) [BBS25] both aim to decompose a network’s parameters into a set of components that are faithful to the original weights, minimal in the number required per input, and individually as simple as possible.

To formalise this, consider a neural network $f(x, W)$ with weight matrices $W = \{W^1, \dots, W^L\}$. A set of parameter components $\{W_c^l\}_{c=1}^C$ are said to comprise the network’s mechanisms if they satisfy the following properties:

- **Faithfulness:** The parameter components should sum to the parameters of the original network.
- **Minimality:** As few parameter components as possible should be required to replicate the network’s output on any given input.
- **Simplicity:** Each component should span as few weight matrices and as few ranks as possible.

APD [Bra+25] approaches this by decomposing the network into a set of parameter components without constraining their rank, using a gradient-based attribution that selects the top- k components most responsible for any given input. While effective on small models, APD scales poorly: its memory cost grows with the number of components, as each component has the same dimensionality as the original network, and its top- k attribution is sensitive to hyperparameter choice.

SPD [BBS25] addresses these limitations by enforcing a rank-one structure on each component, allowing more components to be learned at fixed memory cost, and replacing top- k attribution with a learned causal importance function. Formally, each weight matrix W^l is decomposed into a sum of C rank-one subcomponents:

$$W_{i,j}^l \approx \sum_{c=1}^C U_{c,i}^l V_{c,j}^l, \quad \text{where } U^l \in \mathbb{R}^{C \times d_{\text{out}}}, V^l \in \mathbb{R}^{C \times d_{\text{in}}} \quad (1)$$

where C is a hyperparameter, for the number of components for each layer, l indexes the layer and i, j are the hidden indices. The subcomponent count, C , may be chosen to be larger than both d_{in} and d_{out} , enabling the method to learn features in superposition [BBS25].

To enforce the properties above, three losses are introduced, which are optimised jointly. Faithfulness is enforced through a mean squared error loss between the original network’s weights and the the sum of subcomponents:

$$\mathcal{L}_{\text{faithfulness}} = \frac{1}{N} \sum_{l=1}^L \sum_{i,j} \left(W_{i,j}^l - \sum_{c=1}^C U_{i,c}^l V_{c,j}^l \right)^2 \quad (2)$$

where N is the total parameter count in the target model.

Optimization for simplicity and minimality is achieved through a stochastic masking procedure. Intuitively, if a sub-component is not necessary for reconstruction, it can be sampled to zero without loss of faithfulness. A causal importance function $\Gamma : \mathcal{X} \rightarrow [0, 1]^{C \times L}$ is trained to learn the lower bound of the sampling distribution. In SPD, Γ is a set of independent MLPs γ_c^l that each take a layer’s activation and output a scalar importance for the corresponding subcomponent:

$$g_c^l(x) = \Gamma_c^l(x) = \sigma_H(\gamma_c^l(h_c^l(x))), \quad h_c^l(x) = \vec{V}_c^{l\top} \vec{a}^l(x) \quad (3)$$

where σ_H is a hard sigmoid and $\vec{a}^l(x)$ is the activation at layer l . These importance values serve as the lower bound for how much a subcomponent can be masked. With $r_c^l \sim \mathcal{U}(0, 1)$ sampled per subcomponent, the masked weight matrix is given by:

$$\begin{aligned} m_c^l(x, r) &= g_c^l(x) + (1 - g_c^l(x)) r_c^l, \\ W^l(x, r) &= \sum_{c=1}^C \vec{U}_{i,c}^l m_c^l(x, r) \vec{V}_{c,j}^l, \end{aligned} \quad (4)$$

These masked matrices are trained to reconstruct the original network’s output.

$$f(x \mid \{W^l(x, r)\}_{l=1}^L) \approx f(x \mid \{W^l\}_{l=1}^L). \quad (5)$$

This is done through both a layer-wise and a global reconstruction loss. The layer-wise version is necessary to provide a stronger training signal as randomly masking all components at the same time can be too noisy to learn from.

$$\begin{aligned} \mathcal{L}_{\text{stochastic-recon}} &= \frac{1}{S} \sum_{s=1}^S D(f(x \mid W^l(x, r^s)), f(x \mid W)), \\ \mathcal{L}_{\text{stochastic-recon-layer}} &= \frac{1}{LS} \sum_{l=1}^L \sum_{s=1}^S D(f(x \mid W^1, \dots, W'^l(x, r^{(s)}), \dots, W^L), f(x \mid W)). \end{aligned} \quad (6)$$

where D is a task-specific divergence measure, such as MSE or KL divergence.

Lastly, to enforce minimality, the importance function is trained to activate only the subcomponents needed for reconstruction. The loss balances reconstruction accuracy against the number of active subcomponents, creating a tradeoff between faithfulness and minimality. The goal is to activate only the subcomponents required to reproduce the target model’s output while leaving the rest inactive.

$$\mathcal{L}_{\text{minimality}} = \sum_{l=1}^L \sum_{c=1}^C (g_c^l(x))^p, \quad p > 0. \quad (7)$$

The full SPD objective is a weighted sum of these terms:

$$\mathcal{L}_{\text{SPD}} = \mathcal{L}_{\text{faithfulness}} + \beta_1 \mathcal{L}_{\text{stochastic-recon}} + \beta_2 \mathcal{L}_{\text{stochastic-recon-layer}} + \beta_3 \mathcal{L}_{\text{minimality}}. \quad (8)$$

3.2 Irreducibility and the natural unit of decomposition

To appropriately evaluate the performance of a multi-dimensional feature decomposition method, it’s necessary to have a clear definition of what the ground-truth decomposition is. The original superposition hypothesis states that the hidden states can be represented as a linear combination of sparse one-dimensional features. [Elh+22; Eng+25]

Definition: The hidden state $x_{i,l}$ can be written as:

$$x_{i,l} = \sum_i v_i f_i(t) \quad (9)$$

where $v_i \in \mathbb{R}^d$ are pairwise δ -orthogonal directions ⁴ and $f_i(t) \in \mathbb{R}$ are sparse scalar activations, with $f_i(t) = 0$ outside the domain of feature i .

Recent work has shown, that this may be insufficiently strict [Smi24; Eng+25], as it does not capture the structure of multi-dimensional features. In particular, transformer models are shown to organise cyclical concepts in low-dimensional subspaces that do not decompose into meaningful linear components, but are instead manipulated jointly [Eng+25].

To accomodate these features, the same paper proposes a correction to the superposition hypothesis, alongside a formal definition of irreducibility that defines, when a feature is in its atomic form and when it can be decomposed into smaller units without loss of information.

Definition: A feature $f : \mathbb{R}^i \rightarrow \mathbb{R}^{d_f}$ is truly multi-dimensional, if it cannot be written as a sum of two independent features or as the sum of two non-co-occurring features. More formally, a feature is reducible into features a and b if there exists an affine transformation

$$f \mapsto Rf + c \equiv \begin{pmatrix} a \\ b \end{pmatrix} \quad (10)$$

for some orthonormal $d_f \times d_f$ matrix R , and additive constant c , such that the joint distribution $p(a, b)$ satisfies one of two conditions:

- **p is separable:** $p(a, b) = p(a) p(b)$
- **p is a mixture of two disjoint distributions:** $p(a, b) = w p(a) \delta(b) + (1 - w) p'(a, b | b \neq 0)$,

Here a and b are the first k and remaining $d_f - k$ components of $Rf + c$, p is conditioned on inputs where f is active, $w \in (0, 1)$ and δ is the Dirac delta ⁵. In the mixture, the two distributions have disjoint support, with one distribution having no mass where b is active, and the other having no mass where b is zero.

Under this decomposition, the superposition hypothesis takes the form:

Definition: Every hidden state $x_{i,l}(t)$ can be written as a sum of feature contributions

$$x_{i,l}(t) = \sum_i V_i f_i(t) \quad (11)$$

where each $f_i(t) \in \mathbb{R}^{d_{f_i}}$ is an irreducible feature with internal dimensionality $d_{f_i} \geq 1$, each $V_i \in \mathbb{R}^{d \times d_{f_i}}$ and $\|V_i^\top V_j\| < \delta$ for $i \neq j$.

Secondly, and more importantly for SPD, if the multidimensional superposition hypothesis holds, some features f may require multiple rank-one subcomponents. It is not trivial that an irreducible function f (e.g. as shown in Section ??) arises from sampling two dependent and non-overlapping features that can still be compressed into a single rank-one component if the model learns to group them along a shared direction.

However, when the model learns genuinely multi-dimensional features, these cannot in general be represented by a rank-one subcomponent. Instead, they occupy structured higher-dimensional subspaces, implying either the need for higher-rank representations or post hoc clustering of the underlying rank-one components.

3.3 Rank- k subcomponents and relational importance

In the original SPD paper, the subcomponents are constructed as rank-one matrices, through an outer product of two vectors. This formulation is however incompatible with the multi-dimensional superposition hypothesis. (Section 3.2) A decomposition into lower-dimensional subcomponents, necessarily forces features to be split across multiple subcomponents, or to be approximated through a single subcomponent at the cost of reconstruction.

To represent multi-dimensional features, the pairs of vectors are replaced with pairs of matrices.

$$W_{i,j}^l \approx \sum_{c=1}^C \sum_{r=1}^k U_{c,i,r}^l V_{c,j,r}^l, \quad \text{where } U^l \in \mathbb{R}^{C \times d_{\text{out}} \times k}, V^l \in \mathbb{R}^{C \times d_{\text{in}} \times k} \quad (12)$$

Each subcomponent $U_c^l V_c^{l\top} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ has rank at most k , recovering the original SPD definition when $k = 1$. As in the multi-dimensional superposition hypothesis, k is an upper bound on each subcomponent's representational capacity; the effective rank may be lower,

⁴should i say what delta is or is it clear that its small?

⁵can i get away with just writing $p(a|b = 0)$

In SPD, each subcomponent’s importance is computed independently through an MLP followed by a sigmoid non-linearity, allowing multiple subcomponents to be active for a given input and thus distribute a feature across them, even when it is irreducible. Likewise, the minimality loss does not translate to the rank-k setting, where each subcomponent can represent multiple feature directions: it imposes no penalty for distributing a feature across multiple subcomponents, nor for a single subcomponent to encode parts of multiple features. (Section 3.1)

To address this, the causal importance function is redefined to maximize the conditional probability, that a given subcomponent W_c^l , is responsible for a given feature, conditioned on the activations of all other subcomponents in the same layer.

$$g_c^l(x) = p(W_c^l \mid \{h_{c'}^l(x)\}_{c'=1}^C), \quad h_c^l(x) = V_c^{l\top} \bar{a}^l(x) \in \mathbb{R}^k \quad (13)$$

where $\bar{a}^l(x)$ is the activation of layer l and C is the set of subcomponents in layer l . The distribution is computed with a classic transformer block operating over subcomponents, with a softmax over their activations. [Vas+23]

This produces a normalized assignment over subcomponents, $\sum_{c=1}^C g_c^l(x) = 1$, causing the original minimality loss to become degenerate, as each layer contributes a constant value independent of the model state. Instead, it’s replaced with entropy, to pressure the assignment to concentrate on as few subcomponents as possible.

$$\mathcal{L}_{\text{minimality}} = -\frac{1}{|L|} \sum_{l=1}^L \sum_{c=1}^C g_c^l(x) \log g_c^l(x) \quad (14)$$

An unfortunate limitation of this formulation is its computational cost. Computing relational importance via a transformer over subcomponents scales quadratically with the number of subcomponents compared to the linear cost of the original independent importance functions.⁶

4 Results

4.1 Overview of experiments

To evaluate how the proposed changes affect SPD’s performance, and how these relate to the multi-dimensional superposition hypothesis, the result section is structured around three experiments⁷, each with the purpose of isolating a specific structural property. For an overview see table 1.

The first experiment reuses the Toy model of Superposition (TMS) [Elh+22] once sampled with independent features and once with paired features, where these features are sampled in pairs, creating the simplest case of statistical dependence. In particular, the model learns to group the paired features along a single direction, allowing rank 1 subcomponents to capture full features, without having to split them across multiple subcomponents.

The second experiment also reuses a known experiment from SPD, the Toy model of compressed computation, where a model must learn to compute more functions, than the number of available directions in a lower dimensional space. Particularly, this experiment is interesting to explore how softmax and attention affect the decomposition. Softmax in particular, is best when the ground truth decomposition is meant to pick a singular subcomponent for each feature, due to its properties[some source], the toy model of compressed computation, does not allow for a one-hot assignment, as no disentanglement is possible, and to stay faithful to the original model, should learn to distribute mass across multiple subcomponents.

The third experiment generalizes the paired setting to a k-dimensional case by computing the volume of a simplex from its vertices. The ground truth decomposition is still shown to be dependent, but the features can now not be contained in a single rank-one subcomponent, as the geometry of the feature is more complex. This experiment serves to test how well softmax and sigmoid gating can group feature parts together, when their expressiveness is increased.

4.2 Toy Model of Superposition

The Toy Model of Superposition (TMS) [Elh+22] is trained to reconstruct sparse feature activations through a lower-dimensional bottleneck that forces the model to learn features in superposition. The architecture is a single-hidden-layer autoencoder with tied weights:

$$\hat{x} = \text{ReLU}(W^\top Wx + b), \quad W \in \mathbb{R}^{d_{\text{hidden}} \times d_{\text{features}}}, \quad d_{\text{features}} \gg d_{\text{hidden}}. \quad (15)$$

⁶Play with attention approximations

⁷maybe 4

	Toy Model of Superposition	Toy Model of Dependent Superposition	Toy Model of Compressed Computation	Toy Model of Geometry
Objective	$x \in \mathbb{R} \mapsto x$	$x \in \mathbb{R}^d \mapsto x$	$x \in \mathbb{R} \mapsto x + \text{ReLU}(x)$	$x \in \mathbb{R}^d \mapsto \text{vol}(x) \in \mathbb{R}^+$
Statistical dependence	$\times$	$\checkmark$	$\times$	$\checkmark$
Formal feature dimension	1	2	1	$n(n+1)$
Component rank required	1	1	1	k
Ground-truth decomposition	One-hot	One-hot per pair	Distributed	One-hot per simplex

Table 1: Overview of the toy models and their properties.

TMS is studied in two regimes: one with independent features, reproducing the original SPD setup, and one with paired features that introduces statistical dependence. For both regimes, the rank of each subcomponent is kept at 1, both because this is the ground-truth structure and to isolate the effect of softmax-and-attention.

To quantify how well the learned subcomponents recover the target model, two metrics are computed across both regimes. The mean max cosine similarity (MMCS) [Cun+23a] measures directional alignment between each target feature and its best-matching learned subcomponent:

$$\text{MMCS} = \frac{1}{d_{\text{features}}} \sum_{j=1}^{d_{\text{features}}} \max_c \left(\frac{\vec{U}_{:,c} V_{c,j} \cdot W_{:,j}}{\|\vec{U}_{:,c} V_{c,j}\|_2 \|W_{:,j}\|_2} \right). \quad (16)$$

where $c \in C$ is the parameter component index and $j \in \{1, \dots, d_{\text{features}}\}$ is the input feature index. A value of 1 indicates that every target feature has a perfectly aligned subcomponent.

Apart from directional alignment, the strength of the alignment is also measured, to detect feature shrinkage, a problem apparent in both Sparse Autoencoders and Attribution-Based Parameter Decomposition [JT24; WS24; Bra+25]. The mean L_2 ratio (ML2R) compares the euclidean norm of the best-aligned subcomponent to the target feature [BBS25]:

$$\text{ML2R} = \frac{1}{d_{\text{features}}} \sum_{j=1}^{d_{\text{features}}} \frac{\|\vec{U}_{:, \text{mcs}(j)} \vec{V}_{\text{mcs}(j), j}\|_2}{\|W_{:, j}\|_2} \quad (17)$$

where $\text{mcs}(j)$ is the index of the most aligned subcomponent for feature j . A ratio near 1 indicates the feature is fully captured by its best-aligned subcomponent.

4.2.1 Independent features

Each feature is drawn independently from a uniform distribution over $[0, 1]$. As each feature is one-dimensional and independent of the others, the ground-truth decomposition is a set of d_{features} rank-one matrices, each non-zero only along the c -th column $W_{:,c}$ of the target weights.

Like in the original SPD paper, two TMS models are trained: TMS₅₋₂ with $d_{\text{features}} = 5$ and $d_{\text{hidden}} = 2$, and TMS₄₀₋₁₀ with $d_{\text{features}} = 40$ and $d_{\text{hidden}} = 10$ with an additional identity layer.

SPD achieves near perfect alignment on both models, with MMCS and ML2R approximately equal to 1. The softmax and attention variant, while achieving near-zero losses, does not manage to perfectly align its components with the target model nor concentrate the entire feature magnitude within a single subcomponent, resulting in a lower ML2R with high variance. For the larger model this skews the geometry further.

Insert a table for this!⁸

While the softmax variant does not recover the exact geometry, it correctly concentrates the entire magnitude on the right number of subcomponents, with the rest pruned to near-zero (Figure 1). In terms of causal importance values,

⁸important

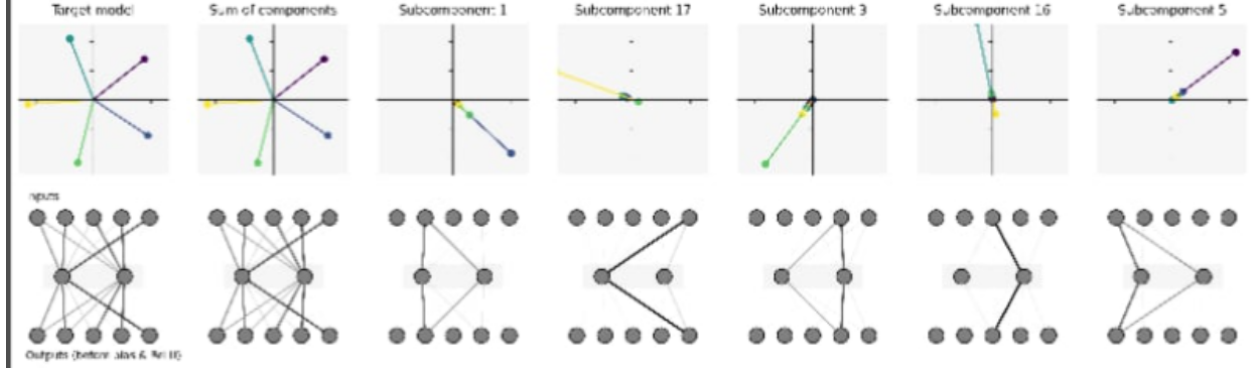


Figure 1: Decomposition of $\text{TMS}_{5,2}$ using Softmax. The first two panels show the target model’s weight columns and the sum of learned subcomponents. Subsequent panels show the individual subcomponents, ordered by decreasing norm. Although 20 subcomponents are allocated, only the first 5 carry meaningful norm and contribute to reconstruction, while the remaining 15 have negligible weight and are effectively pruned by the method. Each active subcomponent recovers primarily one feature direction, with minor leakage across components.

the softmax and original SPD method show the same one-hot assignment for each of the five features in $\text{TMS}_{5,2}$, with some degradation in the larger model where certain subcomponents are causally important for multiple features.

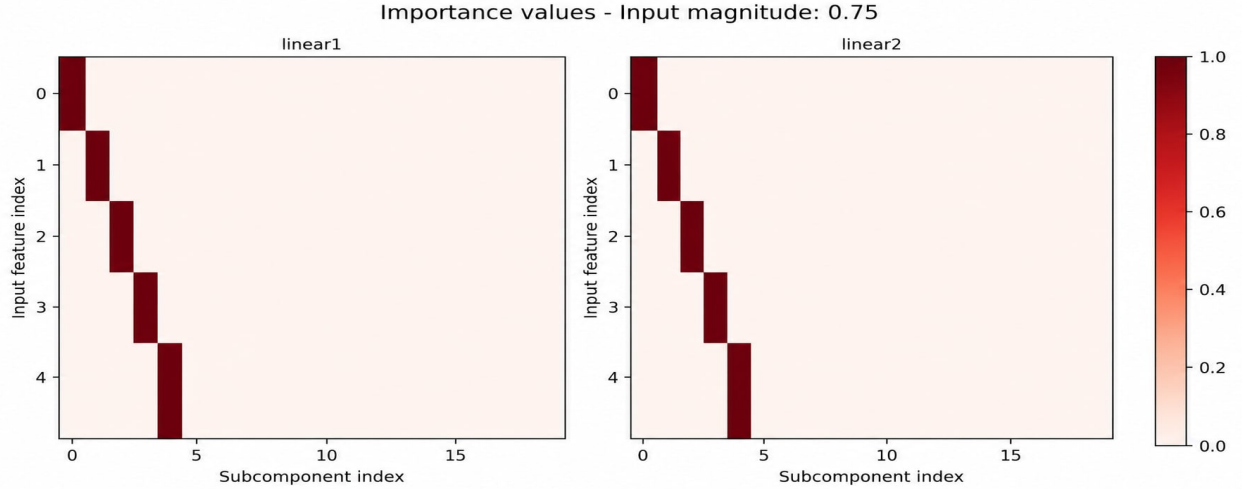


Figure 2: Causal importance values for the softmax variant on $\text{TMS}_{5,2}$, showing a one-hot assignment for each of the five features.

4.2.2 Paired features

In this regime, features are sampled in pairs to create the simplest form of dependence. Features are partitioned into pairs $\{f_{2i-1}, f_{2i}\}_{i=1}^{20}$, with each pair zero with probability $1-p$, and otherwise each pair member drawn independently from $\mathcal{U}(\epsilon, 1)$. The setup uses $d_{\text{features}} = 40$ and $d_{\text{hidden}} = 10$.

Each input is 2-dimensional at construction, but the trained model learns a feature mapping that aligns both pair members along a single direction, with the two features in each pair being manipulated jointly. (Table 2), with within-group cosine similarity averaging 0.999 and across-group similarity averaging 0.053. The pair’s hidden representation therefore collapses to a 1D subspace. Due to this lower-dimensional mapping, paired features can be captured with a single rank-1 subcomponent, without needing to split them across multiple subcomponents.

Group	Mean	Std Dev	Range
Within-group	0.999	0.010	[0.997, 1.000]
Across-group	0.053	0.223	[-1.000, 0.038]

Table 2: Cosine similarity between encoder columns $W_{:,i}$ and $W_{:,j}$, partitioned by group membership.

The ground-truth decomposition is therefore 20 rank-1 subcomponents, with feature indices $\{2i - 1, 2i\}$ assigned to subcomponent i , where the causal importance function should assign all importance to the correct subcomponent for each feature pair, and zero importance to the rest.

In terms of losses, both SPD and Softmax achieve total losses around 10^{-6} , with no significant differences between the two methods. In terms of MMCS, both methods achieve near-perfect alignment, with values around 0.999.

There needs to be another table here.

However, the softmax variant shows a much lower ML2R with high variance, indicating that it struggles to capture the full magnitude of the features within single subcomponents. In terms of causal importances, the two methods show very different patterns, despite identical losses and MMCS. SPD correctly identifies that features should be paired in single subcomponents but does not isolate each pair into its own subcomponent, with several subcomponents carrying importance for multiple feature pairs. The softmax variation correctly assigns each feature pair to a single subcomponent, with the exception of one pair (Figure 3).

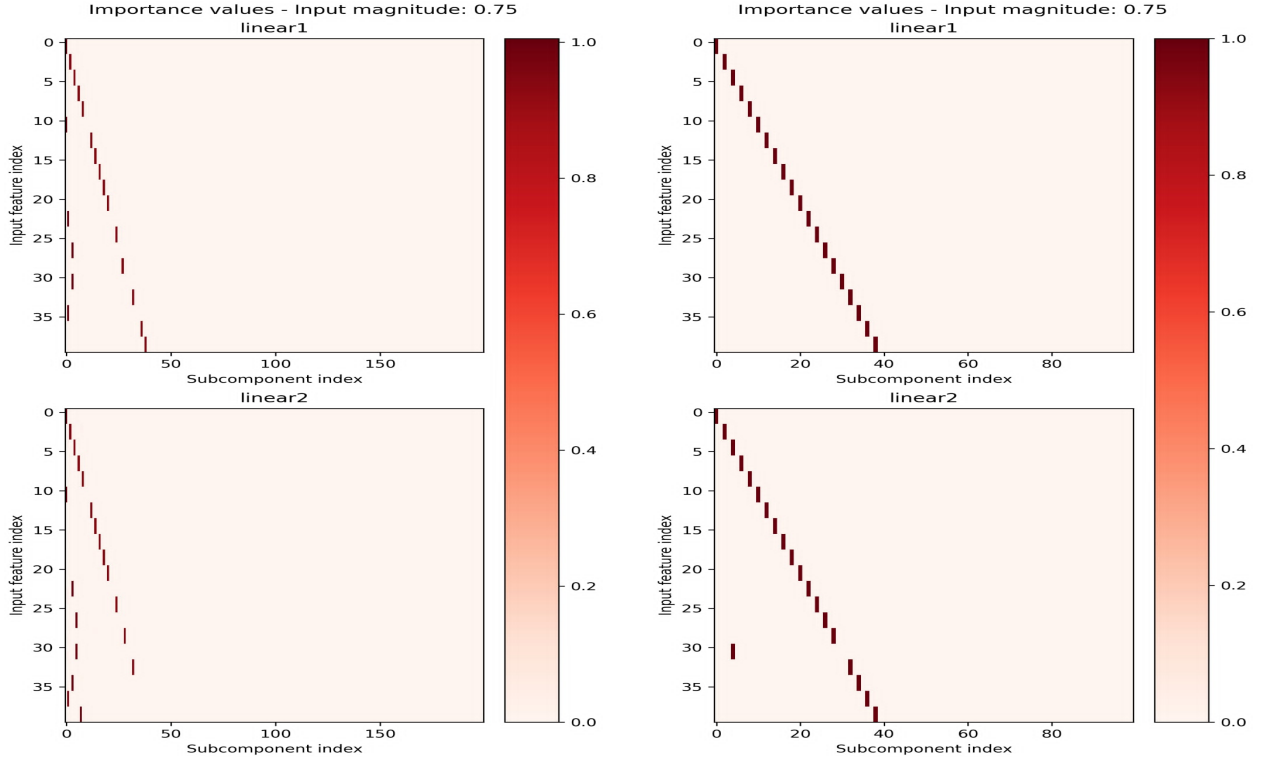


Figure 3: Left: SPD decomposition of the paired TMS model. Right: Softmax of the paired TMS model. SPD finds the correct structure of pairing features together but does not attribute one feature pair per subcomponent. Softmax achieves the desired importance decomposition, with the exception of one feature pair. SPD was trained with 200 subcomponents and Softmax with 100, as these yielded the best results for each method.

Overall across the three settings, SPD with sigmoid gating produces better results on both MMCS and ML2R, despite the more entangled importance assignments observed in the paired regime (Figure 3). The MMCS gap is generally small, but with substantially larger variation in ML2R.

Referring back to the TMS_{5-2} decomposition shown in Figure 1, the softmax method appears to recover a slightly rotated geometry, with residual feature magnitude distributed across adjacent features. The geometric offset may arise

	TMS ₅₋₂		TMS ₄₀₋₁₀		TMS ₄₀₋₁₀ paired	
	SPD	Softmax	SPD	Softmax	SPD	Softmax
MMCS	1.00	0.985 ± 0.016	1.00	0.812 ± 0.105	0.9997	0.9920 ± 0.003
ML2R	1.00	0.797 ± 0.205	1.055 ± 0.045	0.623 ± 0.231	0.796 ± 0.434	0.732 ± 0.350

Table 3: Decomposition quality across both TMS regimes. SPD refers to the original method [BBS25]; Softmax refers to the relational importance function of Section 3.3. If the standard deviation is not reported, it’s because its smaller than 0.001.

as a compensatory mechanism for this residual leakage, allowing the combined subcomponents to maintain accurate reconstruction despite imperfect feature isolation.

4.3 Toy Model of Compressed Computation

The Toy Model of Compressed Computation, a model that was already studied by SPD [BBS25], APD [Bra+25] and more recently by [Bha+25], is a toy model trained to compute the elementwise function $y_i = x_i + \text{ReLU}(x_i)$ through a residual MLP with a lower-dimensional hidden layer.

The architecture consists of an embedding matrix W_E , an unembedding matrix $W_U = W_E^\top$, and a residual MLP with input projection $W_{\text{in}} \in \mathbb{R}^{50 \times d_{\text{emb}}}$ and output projection $W_{\text{out}} \in \mathbb{R}^{d_{\text{emb}} \times 50}$. The input vectors are sparse activations sampled from a uniform distribution over $[-1, 1]$, with the model trained to minimize the MSE between the output and the target function. The model’s computation can be expressed as:

$$y = W_U(W_E x + W_{\text{out}}(\text{ReLU}(W_{\text{in}} W_E x))).$$

For comparability with previous work, the experiments use $d_{\text{features}} = 100$, $d_{\text{MLP}} = 50$, rank $k = 1$ matrices, $C = 200$ subcomponents and lastly an embedding dimension $d_{\text{emb}} = 1000$.

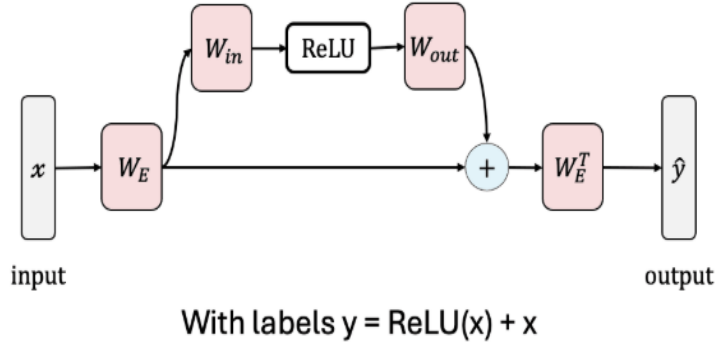


Figure 4: Architecture of the Toy Model of Compressed Computation. Picture taken from [Bha+25]

A naive decomposition of this model would be to let each of the neurons in the hidden layer compute one of the 100 ReLU computations. However this would perform poorly, on the remaining 50 neurons. Instead, to achieve a better reconstruction, the model seems to learn a more efficient representation by assigning multiple features to the same neuron.

The ground truth for W_{in} is clear: each subcomponent should be causally important for one input feature matching the independent one-dimensional structure of the inputs.

For W_{out} , the ground truth is less clear. APD [Bra+25] initially suspected a monosemantic decomposition where each subcomponent corresponds to one output feature, to be the most truthful decomposition. SPD [BBS25] on the other hand reports a polysemantic decomposition in which 50 subcomponents each contribute to multiple features while the remaining are pruned to near-zero norm.

Another paper [Bha+25] supports the interpretation of SPD, showing that the model does not implement 100 separate computations through superposition but rather that they jointly operate within a shared 50-dimensional subspace.

4.3.1 Sampling effect on loss

Before analyzing the decomposition, this experiment will also be used to compare the effect of different sampling distributions on the loss and decomposition quality. Here the two methods SPD and SPD with softmax are compared across two sampling regimes: one-hot sampling, where only one feature is active at a time, and at-least-zero sampling, where each input is independently sampled with probability p .

Softmax generally produces lower losses when sampling is done with one-hot inputs rather than sampling every feature independently. Most noticeably, the Minimality loss, does not converge under at-least-zero sampling. This is unsurprising, with how the loss is defined to maximize the probability of a feature belonging to a single subcomponent. However if the input distribution is not one-hot, the input is a noisy and potentially meaningless combination of unrelated features.

Sigmoid-based importance does not exhibit a significant difference between the two sampling methods, likely since each subcomponent’s importance is evaluated independently, and the loss measures the magnitude of subcomponent activation (Section 3.1) to the correct contribution.

Metric	SPD		Softmax	
	One-hot	At-least-zero	One-hot	At-least-zero
Faithfulness	1.5×10^{-6}	1.4×10^{-6}	3×10^{-5}	5×10^{-5}
Stochastic reconstruction	1.8×10^{-4}	1×10^{-5}	9×10^{-5}	1×10^{-4}
Layerwise stochastic reconstruction	1×10^{-5}	1×10^{-5}	7×10^{-5}	9×10^{-5}
Output reconstruction	3×10^{-6}	4×10^{-6}	8×10^{-5}	1.6×10^{-4}
Minimality	3.83	3.95	8×10^{-5}	4.06

Table 4: Reconstruction quality across sampling regimes for the Residual MLP.

4.3.2 Causal importance

The experimentation with Softmax seems to replicate a similar structure to that found in APD [Bra+25], where one subcomponent is causally important for one feature, while the remaining subcomponents have negligible importance for that feature in both layers.

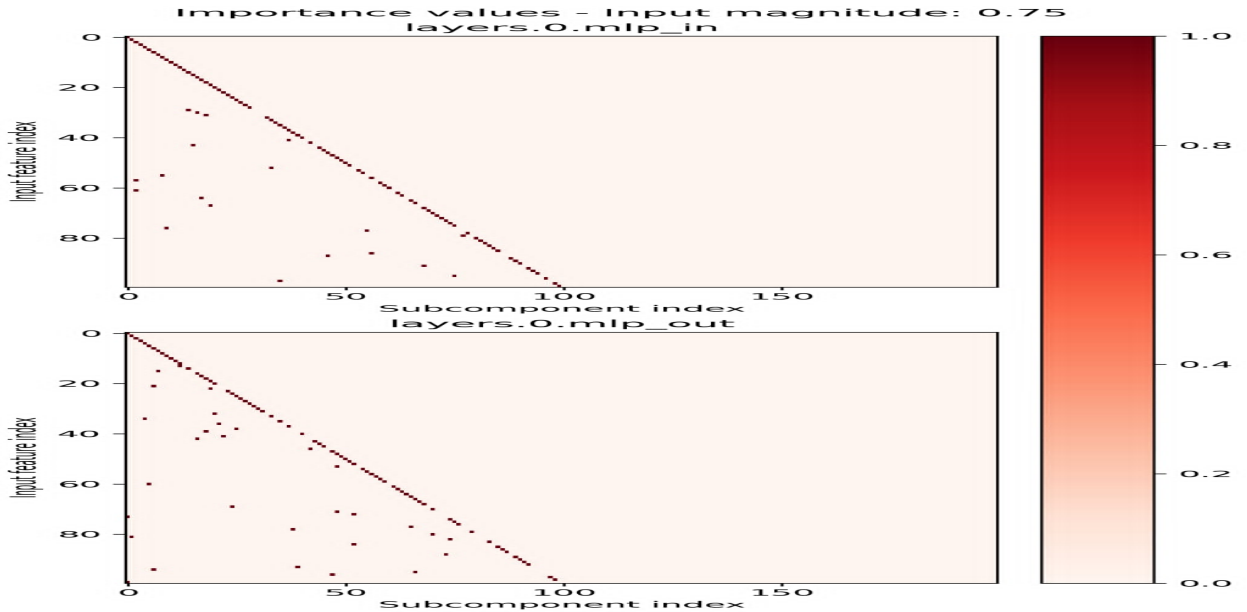


Figure 5: Causal importance values for Softmax on the Toy Model of Compressed Computation. Each input feature activates a near one-hot importance pattern across subcomponents. Sampled using one-hot sampling.

If this reading is taken at face value, there should only be two possible interpretations, either each subcomponent learns a monosemantic representation for one feature, and the remaining subcomponents are unimportant and can be ablated. Alternatively, the minimality loss is too strong and forces the model to concentrate all importance on a single subcomponent, at the cost of reconstruction which should be reflected in a high reconstruction error. Surprisingly, as seen in table 4, Softmax produces near-zero reconstruction error, and should therefore have recovered a monosemantic decomposition.

To examine this, the neuron contributions for both the target model and softmax are plotted in figure 6. The computation of the contribution is computed as in SPD [BBS25].

$$\text{NeuronContribution}_{\text{model}}(i) = (W_{U[i,:]} W_{\text{out}}) \odot (W_{\text{in}} W_E[:, i]) \quad (18)$$

$$\text{NeuronContribution}_{\text{subcomponent}}(i) = \max_{c \in C} [(W_{U[i,:]} U_{\text{out}} V_{\text{out}}^T) \odot (U_{\text{in}}[:, c] V_{\text{in}}^T[c, :] W_E[:, i])] \quad (19)$$

where $W_{E[:, i]}$ and $W_{U[i, :]}$ are the i -th column and row of the embedding and unembedding matrices and $\odot$ denotes elementwise multiplication.

$W_{U[i, :]} W_{\text{out}}$ measures how strongly the neurons contribute to the "reconstruction" of the output, while $W_{\text{in}} W_E[:, i]$ measures how strongly the neurons are activated by the input feature i . If both produce high values, the product will be high, indicating that the neuron contributes strongly to the feature's computation. Otherwise if either is low, the contribution will be low, indicating that the neuron does not contribute significantly to that feature.

As seen in figure 6, Softmax, like SPD, find subcomponents that contribute to multiple features, which also replicates the behaviour of the target model.

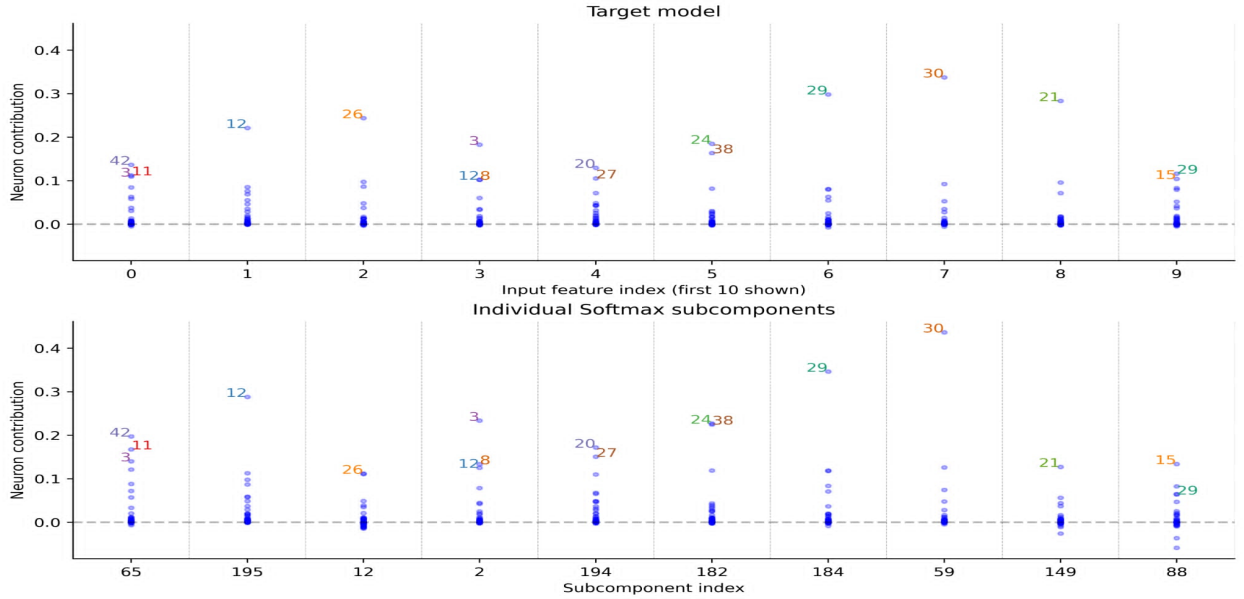


Figure 6: Per-feature contribution of each subcomponent at the neuron level. Subcomponents contribute to multiple features simultaneously, even when their causal importance is concentrated on a single feature.

The one-hot importance assignment is therefore not equivalent to a monosemantic parameter decomposition. One initial reading would be that the subcomponents represent multiple features in superposition, with the observed interference arising from near-orthogonal subcomponents. However, this is contradictory to the results found by [Bha+25], which show that the trained W_{out} operates within a shared 50-dimensional subspace rather than implementing 100 separate computations.

If softmax was genuinely encoding 100 features in superposition, the U_c and V_c vectors of its 100 active subcomponents would have to be near-orthogonal. To examine this, the pairwise cosine similarity between the U_c and V_c vectors of the causally important subcomponents are computed in figure 7.

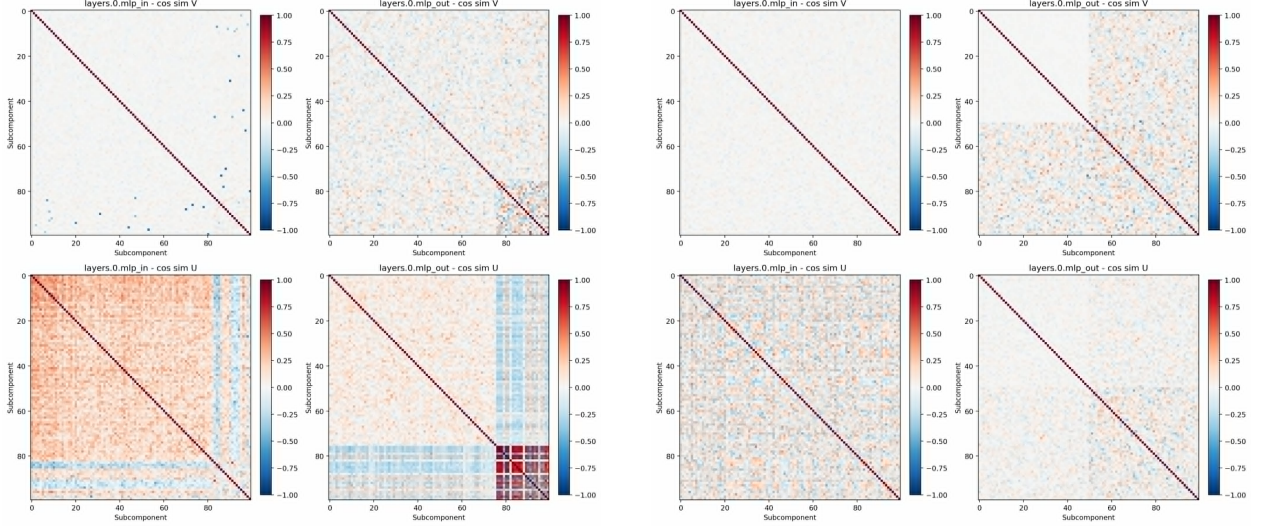


Figure 7: Left: Softmax. Right: SPD. Pairwise cosine similarity between the U_c and V_c vectors of the causally important subcomponents for both layers. Subcomponents are ordered by decreasing norm.

For W_{in} , both methods recover near-orthogonal V vectors, an expected result, as the embedding dimension is large enough to accommodate all pairwise orthogonal input features.

The U vectors, by contrast, must project into the 50-dimensional MLP hidden space and therefore cannot be pairwise orthogonal. Interestingly, the two methods find wildly different projections in the U space. Softmax finds two clusters, where the first 80 subcomponents have large positive cosine similarities with each other and large negative similarities with the remaining 20, while the final 20 are positively correlated among themselves. SPD on the other hand find many subcomponents sharing similar directions, but without the clear two-cluster structure observed in SPD.

For W_{out} , the two methods diverge in their representation. In SPD, the 50 causally important subcomponents recover orthogonal directions in both U and V , while the remaining subcomponents are entangled but carry negligible norm and contribute nothing to the final projection.

The softmax variant, which assigns causal importance to 100 subcomponents, presents a more entangled picture: the first 80 subcomponents have small pairwise cosine similarities in V , while the remaining 20 collapse onto highly similar directions. The U vectors on the otherhand, find the same two-cluster pattern as in W_{in} , with the first 80 subcomponents having small positive cosine similarities with each other and large negative similarities with the remaining 20, while the final 20 practically collapse onto the same direction.

4.4 Subcomponent Norm Distributions

As briefly mentioned in TMS, but also observed in the CC model, softmax produces feature shrinkage, even when it produces the right feature directions. To quantify if this could be the cause of insufficient norm concentration, the norms of the subcomponents are examined. (Figure 8)

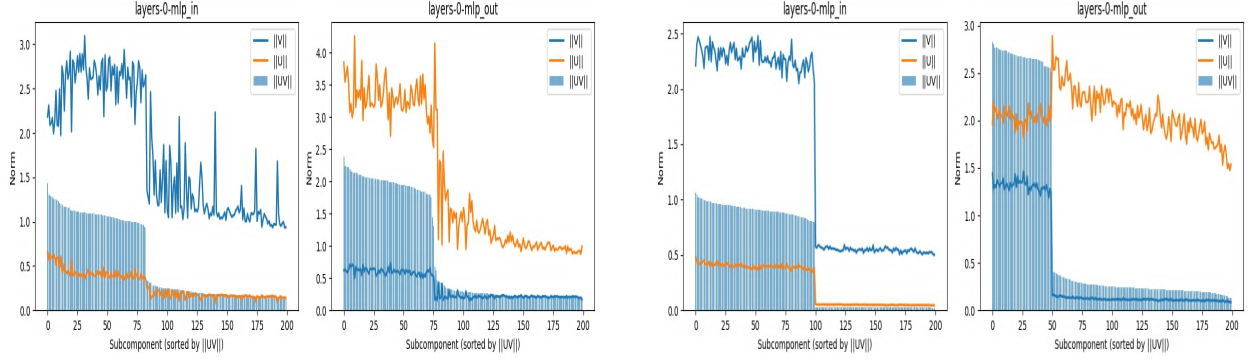


Figure 8: Left: Softmax. Right: SPD. Norms of the subcomponents, ordered by decreasing norm. Matrix norm is the fribenius norm of the subcomponent matrix $U_c V_c^T$. Subcomponents are ordered by decreasing norm.

The two methods recover very different norm distributions. Consistent with the cosine similarity analysis, softmax separates into two broad clusters: approximately the first 80 subcomponents have substantially larger norms, while the remaining 120 retain smaller but non-negligible norms in both W_{in} and W_{out} .

Notably, the softmax does not drive unused subcomponents to zero norm, but rather to a small but non-negligible value.

SPD, by contrast, concentrates norm almost exclusively in the causally important subcomponents: approximately 100 subcomponents for W_{in} and 50 for W_{out} . The remaining subcomponents are effectively pruned, with norms close to zero rather than forming the residual low-norm tail observed in the softmax decomposition.

A further observation is that the norm contributions differ systematically between the two layers. In W_{in} , the norm is dominated primarily by the V vectors, whereas in W_{out} the dominant contribution instead comes from the U vectors. This pattern is consistent across both decomposition methods.

An initial hypothesis for why the softmax formulation fails to produce hard-zero norms was that, unlike the hard-sigmoid gating used in SPD, the softmax distribution cannot produce exact zeros. However, the plot over the causal importance value frequencies rule out this explanation (Figure 9). The learned importance assignments are nearly bimodal, with most concentrated at either 0 or 1, and only a small fraction occupying intermediate values. (Figure 9) The absence of hard-zero norms therefore seems not to arise from the gating distribution itself, but rather from how reconstruction mass is distributed across subcomponents during optimisation. Empirically, this hypothesis was further weakened by replacing the softmax gating with sparsemax [PNM19], which is capable of producing exact zeros in the gating distribution. Despite this, the resulting norm distributions remained qualitatively identical to those obtained with softmax.

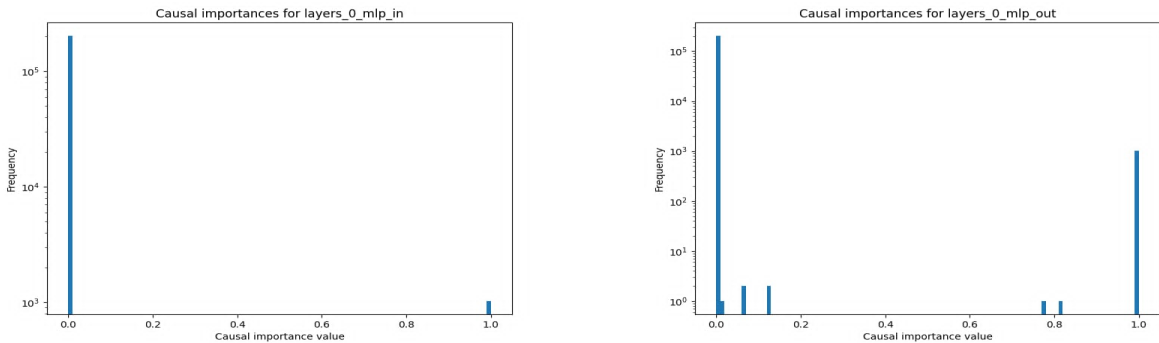


Figure 9: Distributions of causal importance values for the softmax decomposition on both layers.

More generally, this behaviour is likely related to the feature shrinkage phenomenon previously observed in both TMS and CC. The faithfulness objective is computed additively as the MSE between the summed subcomponent

reconstruction and the target weight matrix, making it indifferent to how reconstruction mass is allocated across subcomponents. Consequently, if the importance regularisation only weakly penalises low-magnitude subcomponents, optimisation may converge to solutions in which a small set of dominant subcomponents carry most of the semantic structure, while a long tail of weakly active components absorbs residual reconstruction error.

4.5 Toy Model of Geometry

In the paired TMS regime, even though input features were nominally two-dimensional, the model collapsed them onto a single direction, so a rank-1 decomposition per feature was sufficient. To test the rank- k extension of Section 3.3, a setting is needed where features cannot collapse to a single direction. This experiment generalizes the cube example from the introduction, where the same underlying computation admits multiple valid ground-truth decompositions depending on what each subcomponent is allowed to represent.

The model is trained to compute the volume of an n -dimensional simplex from its $n + 1$ vertices in $\mathbb{R}^n$:

$$\text{Vol}(v_1, \dots, v_{n+1}) = \frac{1}{n!} |\det([v_2 - v_1 \quad v_3 - v_1 \quad \dots \quad v_{n+1} - v_1])|. \quad (20)$$

To mirror the paired-feature setup of Section 4.2.2, the model receives K simplices as input, each sampled with probability p and otherwise set to zero across all of its coordinates:

$$f(x) = W_1 \text{ReLU}(W_2 x), \quad W_2 \in \mathbb{R}^{d_{\text{hidden}} \times K n(n+1)}, \quad W_1 \in \mathbb{R}^{K \times d_{\text{hidden}}}, \quad (21)$$

where $n(n + 1)$ is the total number of coordinates of a simplex in $\mathbb{R}^n$ and K is the number of simplices. Concretely, for this experiment the simplices are triangles in $\mathbb{R}^2$, with $K = 3$ and $d_{\text{hidden}} = 150$.

Unlike TMS, d_{hidden} is chosen to be much greater than the input dimension ($K n(n + 1)$), so that the model does not need to compress features through superposition. The intention is that the model assigns each simplex to a separate subset of neurons, allowing the routing of information from input to output to be studied in isolation as the minimality loss is applied at different strengths and as the rank is increased.

By construction, the $n + 1$ vertices of a non-degenerate simplex are affinely independent: no vertex lies in the affine hull of the others. The volume is therefore a function of all vertices jointly, and no subset can be dropped. Provided the model learns the correct computation, and not some proxy that only approximates the volume, the volume of a single simplex is an irreducible feature in the sense of Section 3.2: it cannot be decomposed further without losing the computation. The volumes of different simplices, in contrast, are separable, as they cover disjoint sets of vertices and can be computed independently.

The ground-truth decomposition should therefore group each simplex into a single unit and separate across simplices. A decomposition method need not recover this, however: depending on its constraints it may instead split a simplex into finer fragments, recovering a single coordinate, a vertex, or an edge rather than the whole simplex. Which of these is recovered should depend on two factors: the dimensionality each subcomponent is allowed to represent, and how strongly the minimality loss penalises spreading importance across multiple subcomponents. For the softmax variant in particular, the entropy minimality loss maximises the probability of a simplex belonging to a single subcomponent. Given sufficient rank and sufficient minimality pressure, the decomposition is then expected to recover a whole simplex as a single component rather than splitting it across several.

4.5.1 Target model structure

While it should be possible for the model to assign a disjoint set of neurons to each simplex, and route their information through disjoint hidden neurons to the output, this is not guaranteed to be the structure that the model learns. To examine this, a similar approach to that of Section 4.3 is adopted.

Concretely, the input features are partitioned into K disjoint groups, where group g contains the $n(n + 1)$ coordinates of simplex g denoted by C_g , with the groups jointly covering all input columns. Let $W_2^{(g)}$ denote W_2 restricted to the columns in C_g and zero elsewhere, so that $W_2 = \sum_{g=1}^K W_2^{(g)}$. The preactivation of hidden neuron i then decomposes additively over the groups:

$$\sum_c W_{2[i,c]} = \sum_{g=1}^K \rho_{i,g}, \quad \rho_{i,g} := \sum_c W_{2[i,c]}^{(g)}, \quad (22)$$

here $\rho_{i,g}$ is the influence of simplex g on neuron i .

On the output side, the strength with which hidden neuron i contributes to the volume computation of simplex k is given directly by the weight $W_{1[k,i]}$.

To measure the routing strength of the path from simplex g to output k through neuron i , the product $\rho_{i,g} \cdot W_{1[k,i]}$ is computed (Adopted from Section 4.3). A large value indicates a strong path, while a small value is less influential. Let s_i denote the strongest of such paths for neuron i across all simplices and outputs.

$$s_i = \max_{g,k} \rho_{i,g} W_{1[k,i]} \cdot^9 \quad (23)$$

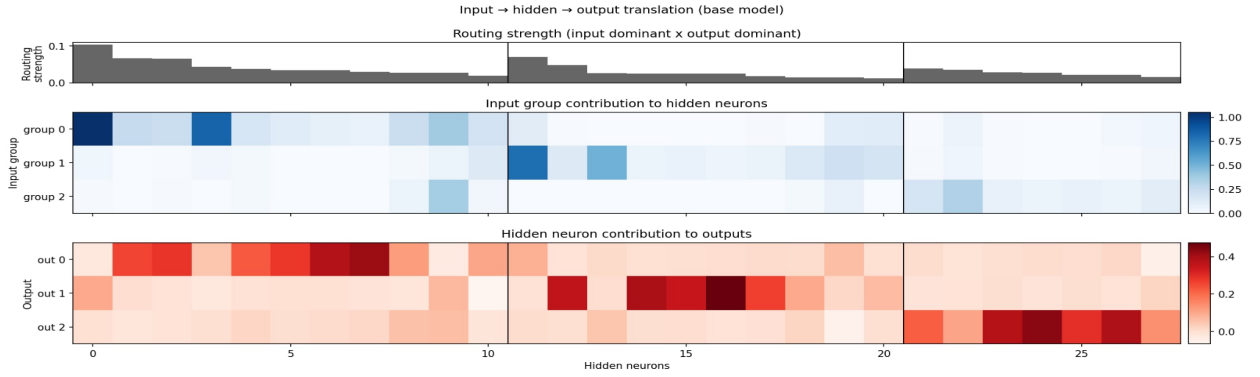


Figure 10: Top: values of s_i . Middle: response $\rho_{i,g}$ from each input group to hidden neuron i . Bottom: contribution $W_{1[k,i]}$ of each hidden neuron to each simplex’s volume output. Neurons are ordered by ρ , then s_i , and lastly by output strength. Neurons with $s_i > 0.05$ shown.

10

11

The model learns a predominantly modular structure, in which each hidden neuron is strongly influenced by only one simplex and contributes strongly to that same simplex’s output (Figure 10). A summarised view of this routing is shown in Figure 11, where the per group values are aggregated across all hidden neurons, and passed through a ReLU. The response of output k to simplex g is

$$r_{g,k} = \sum_i^{d_{\text{hidden}}} \text{ReLU}(\rho_{i,g}) W_{1[k,i]} \cdot \quad (24)$$

⁹Make sure i compute max

¹¹Consider not taking absolute values here or verify that the code doesn't. Also the plot caps at zero, there must be a max somewhere.

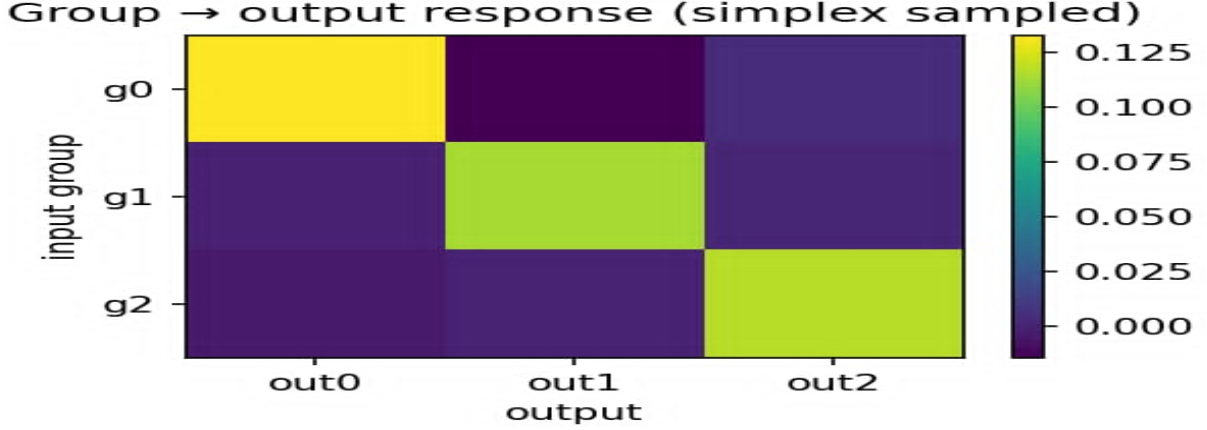


Figure 11: Group to output response $r_{g,k}$, obtained by sampling each simplex in isolation and measuring the resulting volume output. The diagonal structure shows that each simplex predominantly influences its own output, with negligible influence on the others.

The magnitude is concentrated on the diagonal, showing that each simplex strongly influences its own output, with negligible cross-activation. It should therefore be possible to recover a decomposition, where the flow from each simplex is routed through a disjoint set of hidden neurons to its corresponding output, with minimal leakage across subcomponents.

To isolate the effect of the subcomponent rank from that of the minimality coefficient, the two factors are varied one at a time, following a one-factor-at-a-time approach. First, the rank is fixed at $k = 1$, recovering the original SPD formulation, and the minimality coefficient β_3 is varied to examine its influence on the decomposition. Second, β_3 is fixed at a value that produced a clear decomposition at $k = 1$, and the rank is increased to examine how higher k allows different decompositions to be recovered. While a full sweep over both factors was carried out, the one-factor-at-a-time presentation is sufficient here: once the minimality coefficient reached its effective range, the resulting decompositions were qualitatively the best across all ranks.

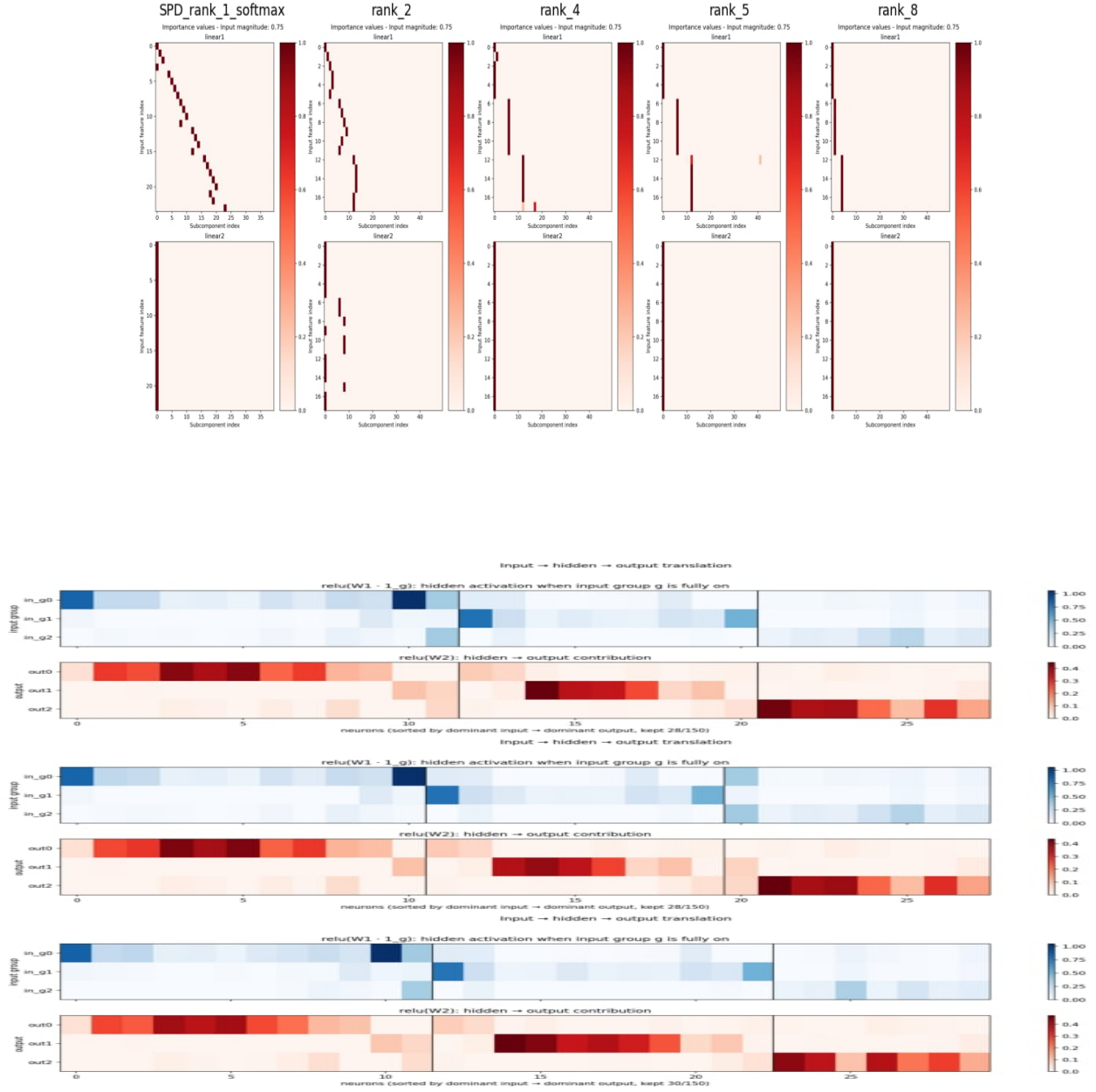
4.5.2 Comparison of importance values across different minimality coefficients

With the target model’s modular structure established, the first sweep fixes the rank at $k = 1$ and varies the minimality coefficient β_3 , comparing how the original SPD gating and the softmax variant allocate importance as the pressure toward concentration increases. At $k = 1$, neither method can represent a full triangle within a single subcomponent, since a triangle occupies more than one dimension; the question is therefore not whether a simplex is recovered whole, but how the two methods distribute a simplex across the rank-1 subcomponents available to them, and how that distribution responds to β_3 .

A low minimality coefficient places little pressure on the importance assignment, allowing a simplex to be spread across many subcomponents. As β_3 increases, the assignment is pushed toward fewer active subcomponents per simplex. The two methods are expected to differ in how this concentration proceeds: [CONFIRM with results — e.g. SPD concentrates only within a narrow band of β_3 and otherwise either over-splits or collapses, whereas the softmax variant concentrates more smoothly across the range].

4.5.3 Comparison of importance values across different ranks for fixed minimality.

mention if it affects losses.



We make a new plot, where for each group we just take the largest responding ones, that way for each group the same neuron can be in all plots.

4.6 Computing

Ideally after constructing these rank k components, then it should be feasible to fully reconstruct the output of the model, by summing these causally important subcomponents together, without any loss of information. In this case the rank k component should fully capture it.

We try this only for Layer 1 and layer 2 first, see if they replicate the layer output.

We do the input, output plot, where we only probe the causally important subcomponents.

5 Discussion

The original spd method produces hard zeroes, something that softmax cannot do. Entmax didnt work either nor did sparsemax, however non hard zeroes do cause magnitude to leak even if marginal, which may increase with more components.. important to mention that the str uture that spd decomposes into is not clearsometimes one hots ome.

Talk about how its unclear what structure spd wants. Even rank 1 is not the smallest possible, as we can align two things to the same direction, but these do not split.

Relate back to the interesting thing that one feature per component and rank 1 matrices does in fact not mean the smallest possible decomposition, as we have shown with TMSs and TMS paired, one is in a way strictly larger, but they result in the same thing!

Talk about how softmax if working corretly, is probably better at extracting relevant features, not how it decomposes of smaller parts. Here it could be nice to touch on, when there genuinely is a one hot solution contra not, i.e in a lower dimnsional subspace.

Thesis needs to discuss how many of the scenarios from the irreducibility section are actually tested -Tms serves as dependent features that cannot be split, -CC serves as an independnet feature - idk if we can do sometiing with wout in this sense?

-Most likely conclusion becomes that we only discuss cases, where the ground truth is dependence not if it manages to split non irreducible features.

Discussion needs to engage with the point of, should the importance value try to replicate model behaviour or should it try to find a more disentangled representaiton like SAE's do?

Section that discusses new SPD variant and how it addresses some of the issues with minimality.

6 Discussion

6.1 Summary of empirical findings

- **TMS, independent features:** SPD attains higher MMCS and ML2R; the softmax variant recovers the correct number of active subcomponents but with magnitude shrinkage. Both reach near one-hot importance.
- **TMS, paired features:** both reconstruct faithfully, but the softmax variant assigns one subcomponent per pair while SPD spreads each pair across several subcomponents. SPD still leads on the directional metrics.
- **Compressed computation:** near one-hot importance does not correspond to a monosemantic decomposition; the 100-subcomponent, 50-dimensional geometry forces shared output directions. Consistent with the 50-dimensional subspace interpretation [Bha+25].
- **Geometry (Experiment 3):**
- **Cross-cutting observation:** sparsity in the importance assignment does not imply sparsity in the underlying representation; magnitude leaks across subcomponents independently of the importance values.

6.1.1 Summary of empirical findings

Across the experimental settings, both SPD and spd with softmax gating produced faithful decompositions, that were faithful to the target model's computations. The main difference between the two was not directly reflected in the reconstruction quality, but was left hidden in the structure of the decompositions they produced.

On the Toy Model of Superposition with independent features, the original SPD method attained a similar mean max cosine similarity, but more accurate mean L_2 ratio, recovering both the direction and the magnitude of each feature accurately. The softmax variant recovered the correct number of active subcomponents, pruned the remainder to near-zero, but concentrated less of each feature's magnitude within its best-aligned subcomponent, as there was some leakage across the active subcomponents, producing a lower mean L_2 ratio with higher variance. This came despite the fact that softmax and SPD achieved the same ground-truth importance assignment.

When the input features were sampled in pairs, introducing statistical dependence, both methods reconstructed the target faithfully, but their importance assignments diverged. Softmax assigned each feature pair to a single subcomponent, recovering the grouped structure that the dependence induces, while SPD showed that multiple input features were grouped within the same subcomponent. Despite this cleaner assignment, the softmax variant stil underperformed on the directional alignment and magnitude metrics.

In the Toy Model of Compressed Computation, where a near one-hot importance assignment was observed, the initial interpretation was that the model had learned a monosemantic decomposition, with each subcomponent representing a single feature. However, this interpretation seemed to contradict the results of [Bha+25], which showed that the model’s computation operates within a shared 50-dimensional subspace rather than implementing 100 separate computations. Additionally on close analysis, each subcomponent was associated with multiple features, which contradicts the monosemantic interpretation of the one-hot importance assignment.

On the Toy Model of Geometry, where each feature is a simplex whose volume depends jointly on all of its vertices and so cannot be reduced to a single direction, the recovered decomposition depended on both the subcomponent rank and the strength of the minimality pressure.

Taken together, these findings point to a single recurring observation: sparsity in the importance assignment does not imply sparsity in the underlying representation. Even when importance becomes near one-hot, the magnitude of a feature may remain distributed across multiple subcomponents, since the faithfulness loss is indifferent to how the reconstruction mass is allocated. This distinction between the sparsity of the assignment and the sparsity of the representation organises much of the discussion that follows.

6.2 Sparsity of assignment is not sparsity of representation

- Claim: a sparse / one-hot causal importance assignment does not imply a sparse (monosemantic, norm-concentrated) underlying representation. Importance lives on the assignment side; the representation is a separate object.
- Prior work reads one-hot importance as evidence of a monosemantic decomposition; the CC results show this inference is invalid (one-hot importance, yet shared structure).
- Two senses, computed on different objects:
 - importance is read off $V^\top a$ (the read side),
 - the contribution to a feature comes from $UV^\top$ (the representation),
 - these can and do disagree – cite Fig. 6 (concentrated importance, but subcomponents contribute to multiple features).
- Modus tollens (restate briefly): IF softmax genuinely encoded 100 features in superposition, its active U_c, V_c would be near-orthogonal. They are not (Fig. 7), and W_{out} must project into the 50-dim MLP space; consistent with the shared 50-dim subspace of [Bha+25]. So concentrated importance $\neq$ superposition encoding.
- The non-orthogonality is therefore structural ($C > d_{\text{MLP}}$), not a sign of superposition.
- Same decoupling, measured in norms instead of monosemanticity: tightening the minimality coefficient sharpens the ASSIGNMENT but not the representational MAGNITUDE.
 - softmax leaves a residual low-norm tail / no hard zeros, SPD hard-prunes – Fig. 8; the magnitude cost shows up as low ML2R in Tab. 3.
 - mechanism: the additive faithfulness MSE is indifferent to HOW reconstruction mass is split across subcomponents, so a penalty on g (Sec. 3.1) controls which components are nominally active, not the magnitude they carry.
- Controls ruling out the gating distribution as the cause:
 - importance is already near-bimodal at 0/1 (Fig. 9), so soft gating is not what blocks hard zeros.
 - sparsemax (capable of exact zeros) yields qualitatively identical norm distributions – leak is not from the gating nonlinearity.
- Controlled demonstration (geometry, Sec. ??): sweeping β_3 sharpens the assignment (concentrates onto fewer subcomponents) while magnitude does not fully follow.
- Synthesis (state once): assignment-level pressure (minimality, sparser gating) cannot by itself control representation-level structure – neither monosemanticity (A) nor magnitude (B).
- Implication for practice: importance values must not be used on their own as a measure of representational sparsity, monosemanticity, or norm concentration. A representational check is required – contribution analysis, U/V cosines, norm distribution, effective rank.
- Caveat applying to our own Experiment 3: the rank sweep shows importance concentrating onto $\sim K$ sub-components, but by this argument that is not yet proof the subcomponents OWN whole simplices – needs representational confirmation (U/V structure, effective rank $\approx$ simplex dimensionality).

6.2.1 Sparsity of assignment is not sparsity of representation

A repeating observation across the experiments is that a sparse causal importance assignment does not imply a sparse underlying representation. Particularly importance is a property of the assignment, where the importance is calculated on the input projection of each subcomponent, while the full representation is a separate object, that is determined by the full contribution of each subcomponent UV . The two need not coincide, and in these experiments have disaligned.

The clearest example of this comes from the Toy Model of Compressed Computation. Here, Softmax assigns one-hot importance, which taken in conjunction with the stochastic reconstruction loss, which uses this importance assignment as a lower bound, would make it natural, given that the model achieves near-zero reconstruction error, to read this as evidence of a monosemantic decomposition, where all other subcomponents can be ablated without affecting the model’s behaviour.

However examining the representation contradicts this assumption, every subcomponent has one-hot importance, yet every subcomponent contributes to multiple features (Figure 6). Particularly this both implies that the subcomponent should have importance to multiple features, and not contain all of the mass for a single feature.

Ofcourse, this could be read as evidence of superposition, where feature leakage arises from near-orthogonal subcomponents, but this is contradicted by the results of [Bha+25], which show that the trained W_{out} operates within a shared 50-dimensional subspace rather than implementing 100 separate computations.

This disalignment in the Softmax variant is also observed in the magnitudes, where sharpening the minimality coefficient concentrates importance onto fewer subcomponents, but does not concentrate the entire representational magnitude to the same degree, leaving a residual low-norm tail rather than pruning to hard zeros as SPD does (Figure 8) and is influential in the lower mean L_2 ratio in the Toy Model of Superposition (Table 3). Initially it was thought, that the softmax gating, which cannot produce exact zeros, was responsible for this residual mass, but the near-bimodal importance distribution (Figure 9) and the qualitatively identical norm distribution obtained with sparsemax, which can produce exact zeros, rule out this explanation.

These disalignments are also easily overseen, where the decompositions show clear importance assignments, that look convincing with respect to the expected decomposition, while also achieving good reconstruction and faithfulness metrics.

Faithfulness computes the difference between the summed subcomponent reconstruction and the target weight matrix, so it is indifferent to how reconstruction mass is allocated across subcomponents.

Similarly reconstruction computes the output of the model given all the subcomponents, which again allows multiple equally faithful and equally reconstructive decompositions with different mass allocations.

A penalty on the importance values g (Section 3.1) therefore controls which subcomponents are nominally active, not the magnitude they carry. [This part requires the knowledge from the geometry experiment, where we will see how SPD does when none of its components are active.]

Taken together, the minimality loss in neither its original form nor the softmax variant, can be used on its own to control or even measure the sparsity of the underlying representation, and must currently be supplemented with a separate representational check.

[This is a good bridge to the VSPD paper, where both can be ablated together something?]????

6.3 Relation to adversarial parameter decomposition

- VPD [bushnaq2026interpreting] identifies the same assignment-representation gap from the ablation side: the minimality loss can be satisfied by splitting a mechanism into narrow subcomponents under stochastic masking.
- VPD’s fix is adversarial ablation, which activates supposedly unimportant subcomponents in combination and penalises any residual representation they carry; plus a superlinear frequency penalty.
- The relational importance function attacks the same gap from the assignment side (competition for ownership) rather than the ablation side.
- The two are complementary: adversarial sampling could be combined with relational importance.
- The observed magnitude leakage (small non-zero norms on unimportant subcomponents) is exactly the structure adversarial ablation is designed to remove, pointing to a concrete next step.

6.4 Rank extension and the structure of the recovered weights

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This should also mention that rank k does not work with the minimality loss that SPD uses, as we see that the components just split the same feature across multiple subcomponents, again a consequence of that faithfulness is indifferent to how the reconstruction mass is allocated across subcomponents.

And if not mentioned in the section above, it should mention how the faithfulness even when everything is zero, still has some decomposition? This did however not occur, when spd-baseline was run, perhaps this comes as a consequence of that many features are responsible for one computation, so it can be spread across even more features? Use the causal importance frequency plots

6.5 The non-uniqueness of the minimal decomposition

- Rank-1 with one component per feature is not, in general, the minimal description: in the paired regime, an aligned pair can be captured by one rank-1 subcomponent or split across two, both reconstructing identically.
- The rank constraint therefore does not determine a unique decomposition; it admits a coarser grouping and a finer splitting of the same aligned directions.
- Minimality in the number of components is a property of how the importance function groups or splits, not of the rank constraint alone.
- This is the ambiguity the relational importance function is designed to resolve: pressuring jointly-manipulated directions toward a shared subcomponent.
- Connect to the underdetermination of the objective: reconstruction plus ℓ_p minimality plus rank-1 does not pin down a canonical factorisation.

6.6 The non-uniqueness of the minimal decomposition

A rank-1 decomposition with one component per feature is often taken to be the minimal description of a computation, but the paired regime shows that it is not. When a feature pair is aligned onto a single direction, it can be captured by one rank-1 subcomponent or split across two, and both reconstruct the target identically. The rank constraint therefore does not pick out a unique decomposition: it admits both a coarser grouping and a finer splitting of the same aligned directions, with no difference in faithfulness to separate them. Minimality in the number of components is thus not determined by the rank constraint alone, but by how the importance function chooses to group or split. This is the ambiguity the relational importance function is designed to resolve, pressuring directions that are manipulated jointly toward a shared subcomponent rather than leaving the grouping to chance. More broadly, it shows that reconstruction, ℓ_p minimality, and a rank-1 constraint together do not pin down a canonical factorisation; the decomposition that emerges is selected as much by the inductive bias of the importance function as by any intrinsic property of the model.

6.7 What the experiments test within the irreducibility framework

- TMS paired tests dependent features that should be grouped, not split: the relational variant is evaluated on whether it groups correctly.
- Compressed computation tests an effectively independent feature in a constrained subspace.
- The mixture condition of the irreducibility definition is not directly evaluated; flag as future work.
- The thesis therefore primarily tests whether dependent or irreducible features are correctly grouped, rather than whether reducible features are correctly split.

6.8 Limitations and future work

- The softmax variant does not produce hard zeros; sparsemax and entmax do not resolve the residual magnitude, which may grow with the number of subcomponents.
- The relational importance function scales quadratically in the number of subcomponents, unlike SPD’s linear independent importance functions.
- SPD recovers clean structure on the geometric model only within a narrow minimality range, raising a robustness question.
- Future directions: adversarial ablation combined with relational importance; attention approximations to reduce the quadratic cost; direct evaluation of the mixture condition.
- This paper does not address mixtures, and therefore does not conclude on every aspect of irreducibility.

6.9 Conclusion

7 Acknowledgements

8 Other things to consider?

Minimal descriptive length?

8.1 That some features are multidimensional

Think back to SAE, that learn a dictionary of features, then find the linear combination of those features, however that this fails for cyclical features, as it’s cheaper to learn a single feature, than learn a direction and a periodic function. Now for your experiment, the dream scenario is that we allow a multi dimensional component, and that this can learn the cyclical feature, i.e a weekly cycle or modular arithmetic either do this or extend TMS to be multi-dimensional, show if it works better with our gnn or not, likewise we need to show that through individual masking of features, we can teach a component to kill one of its features, if it is better explained by a rank 1 component than rank 2.

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10 Appendix

Descriptive length thing? + approximation of attention experiment

11 Notes

Putting the minimality loss entropy higher, encourages diagonals? Instead of capturing two features in the same component.

Increasing minimality loss encourages feature grouping.

SAE: learn to decompose activations, i.e the output of a layer, Transcoders: learn to approximate the input-output function of a layer

$$\int p(a, b) da db = w \int p_A(a) da \int \delta(b) db + (1 - w) \iint p_{AB}(a, b) da db = w + (1 - w) = 1$$

to read: Terence Tao, *An Introduction to Measure Theory*: Section 1.1 Prologue: The problem of measure — sets up why measure theory exists and why "lower-dimensional sets have measure zero" is the central idea. This alone might give you the intuition you need. Section 1.2 Lebesgue measure — gives you "measure zero," "Lebesgue measure on $\mathbb{R}^n$," and the fact that lines/planes/etc. have measure zero in higher-dimensional ambient spaces. This is the technical core for your question. Section 2.3 Probability spaces — short section that frames everything in probability language, which is what you actually want.

11.0.1 Hard maybes

11.0.2 Per-dimension ablation and implicit rank selection (Hypothesis: ask lukas)

Following the extension to rank- k subcomponents, and the alternative formulation of the GNAN, the output $g_c^l \in [0, 1]^k$ is a per feature importance value, and thus the masking can be applied independently to each feature. The intuition behind this is that if a mechanism captures a 1 dimensional feature, it would only need of the k dimensions to be active.

Likewise if this actually ends up working, k can be considered an upper bound or a so called "capacity" of a mechanism, it also follows nicely with the descriptive length principle found in the appendix of APD.

To determine the effective rank of each subcomponent, the stochastic masking becomes:

$$[m_c^l(x, r)]_j = [g_c^l(x)]_j + (1 - [g_c^l(x)]_j) r_{c,j}^l, \quad r_{c,j}^l \sim U(0, 1) \quad (25)$$

for each j in $1, \dots, k$

Notably this treats each feature independently, which may not be desirable, as such we could modify it ever so slightly:

$$[m_c^l(x, r)]_j = [g_c^l(x)][g_c^l(x)]_j + (1 - [g_c^l(x)][g_c^l(x)]_j) r_{c,j}^l, \quad r_{c,j}^l \sim U(0, 1) \quad (26)$$

The signal becomes an importance of two factors, if the component is important and both features are important, it remains active, if only one feature is important, the other will be pushed to zero

The masked weight matrix becomes:

$$W'^l(x, r) = \sum_{c=1}^C U_c^l \text{diag}(\mathbf{m}_c^l(x, r)) V_c^{l\top} \quad (27)$$

Where the diagonal matrix is the per-feature mask for subcomponent c .

The importance minimality loss penalises all k dimensions of all subcomponents toward zero:

$$\mathcal{L}_{\text{import}} = \sum_{l=1}^L \sum_{c=1}^C \sum_{j=1}^k |[g_c^l(x)]_j|^p \quad (28)$$

This follows the same logic by which SPD selects the number of active subcomponents: the minimality loss drives unused components toward zero, and only those that are genuinely required to reconstruct the target model's output survive.

This makes k a soft upper bound rather than a sensitive hyperparameter. In practice, k can be set conservatively large and the effective rank of each subcomponent is learned from the data.

12 To add:

<https://arxiv.org/pdf/2303.13506> quantization, multidimensional feature crossover We use the method from the irreducibility paper to quantize how well it is seperable, mixture like. Rewrite irreducibility to mkae it clear that it is the function

The method is quite complex, we practically look for the input space to see where a feature comes from and its dimension but when we look to decompose a layer, we look at f , the projection of the input space to see its rank, but we look for the values that the input space get mapped to. So in other words, the f decides how many subcomponents we need, but the f decides how many features are inside of it?

quantization and spd clustering

why softmax + attention instead of only softmax + minimality.

Somewhere add this whole thing of importance maybe being an interpretability trap? Or that it should be seen as a caution thing, it will predict what is responsible, not the model alignment.

The thing that makes multifeature hypothesis perfect

Multifeature hypothesis says that there exists a transformation f , to a lower dimensional space, in this case any W linear map,

that can be paired with some new V_i that describes the hidden state, in this case it's the SPD components!!!! These perfectly make this up and pair this with quantization and we are golden.

Remark rewrite TMS paired into TMS, they should be the same section. Remember V is the input reading so if w

In general add how each experiment relates to the multi dimensional feature hypothesis decomposition.

Remember to add the VPD to related work, remember to add the citation that irreducibility definition is wrong.

12.0.1 SVD reformulation, capacity maybe per feature ablation TBD

+ add read write streams, they're a general thing - ζ perhaps issues with diffusion and recurrent models where this stream is not as clear. <https://arxiv.org/pdf/2504.00194> + This section must explain delta orthogonality, visualization of Read, write matrices, perhaps nudge the SVD view. Most importantly should address if models learn lower dimensional representations or if it's another k sensitivity as in APD.

The competition pressure introduced by the softmax has a theoretical motivation under the multi-dimensional superposition hypothesis. The hypothesis predicts that different irreducible features occupy approximately orthogonal projection subspaces; the softmax-and-attention mechanism implements a training pressure toward this orthogonality at the level of SPD's atoms. Atoms whose V matrices share input directions cannot be cleanly distinguished on inputs that activate those directions, and the softmax penalises ambiguous allocations. The optimiser's response is to push the atoms' V matrices toward disjoint subspaces, recovering the orthogonality structure the hypothesis posits.

Remember to add an experiment that shows if softmax enforces the orthogonality constraint from engels.

Remember to add the correct mean max cosine similarity and l2 values to the tms section.

13 Lukas questions

- Show him CC, - Show him geom - ζ ideas are that i want to both show how "unimportant" the importance values are, for the final decomposition. no readout is still faithful - Ask about irreducibility - ζ i define mixtures and so fourth, but it's not really used anywhere. Is that okay? It's really hard to assess if it will be one or not. - It will be in my discussion that we don't examine the mixture case directly. "The independence" is technically between input pairs, and within pair is dependent. - Ask about formalities, should i do a query over how i've done literature search, should my appendix contain my hyperparameter strategy, or can i just link wandb reports, with hyperlinks? - Do you want the github repo attached to my handin? - I will exceed the page limit if i use this many images, but that's ok right?

-wants code

-wandb + hyperparameters in appendix.